Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56b)

Metric

$$ds^2 = \left(\frac{8\,\epsilon\kappa_0 q_0^2 r e^x - \left(\kappa_0 q_0^2 e^x - 4\,m_0 r\right)u}{2\,r^2 u}\right)\mathrm{d}u \otimes \mathrm{d}u - \mathrm{d}u \otimes \mathrm{d}r - \mathrm{d}r \otimes \mathrm{d}u + r^2 \mathrm{d}x \otimes \mathrm{d}x + r^2 \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1)\ du$$

$$n = \left(-\frac{2\,\epsilon\kappa_0 q_0^2 e^x}{ru} + \frac{\kappa_0 q_0^2 e^x}{4\,r^2} - \frac{m_0}{r}\right)\ du + (1)\ dr$$

$$m = \left(\frac{1}{2}\sqrt{2}r\right)\ dx + \left(-\frac{1}{2}i\sqrt{2}r\right)\ dy$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}r\right)\ dx + \left(\frac{1}{2}i\sqrt{2}r\right)\ dy$$

Contravariant components

$$l = (-1)\ dr$$

$$n = (-1)\ du + \left(-\frac{2\,\epsilon \kappa_0 q_0^2 e^x}{r u} + \frac{\kappa_0 q_0^2 e^x}{4\,r^2} - \frac{m_0}{r}\right)\ dr$$

$$m = \left(\frac{\sqrt{2}}{2\,r}\right)\ dx + \left(-\frac{i\,\sqrt{2}}{2\,r}\right)\ dy$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,r}\right)\ dx + \left(\frac{i\,\sqrt{2}}{2\,r}\right)\ dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u - \frac{2\,\epsilon \kappa_0 q_0^2 e^x}{r u} + \frac{\kappa_0 q_0^2 e^x}{4\,r^2} - \frac{m_0}{r}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\,r}\,\partial_x - \frac{i\,\sqrt{2}}{2\,r}\,\partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,r}\,\partial_x + \frac{i\,\sqrt{2}}{2\,r}\,\partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{2\,\epsilon\kappa_0 q_0^2 e^x}{r^2 u} + \frac{\kappa_0 q_0^2 e^x}{4\,r^3} - \frac{m_0}{r^2}$$

$$\lambda = 0$$

$$\nu = -\frac{\sqrt{2}\epsilon\kappa_0 q_0^2 e^x}{r^2 u} + \frac{\sqrt{2}\kappa_0 q_0^2 e^x}{8\,r^3}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{\epsilon\kappa_0 q_0^2 e^x}{r^2 u} + \frac{\kappa_0 q_0^2 e^x}{4\,r^3} - \frac{m_0}{2\,r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\kappa_0 q_0^2 e^x}{4\, r^4}$$

$$\Phi_{12} = -\frac{\sqrt{2}\epsilon \kappa_0 q_0^2 e^x}{2\, r^3 u} + \frac{\sqrt{2}\kappa_0 q_0^2 e^x}{8\, r^4}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{\sqrt{2}\epsilon \kappa_0 q_0^2 e^x}{2\, r^3 u} + \frac{\sqrt{2}\kappa_0 q_0^2 e^x}{8\, r^4}$$

$$\Phi_{22} = \frac{2\, \epsilon \kappa_0 q_0^2 e^x}{r^2 u^2} - \frac{\epsilon \kappa_0 q_0^2 e^x}{r^3 u} + \frac{\kappa_0 q_0^2 e^x}{8\, r^4}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{2\, \epsilon \kappa_0 q_0^2 e^x}{r^3 u} + \frac{\kappa_0 q_0^2 e^x}{2\, r^4} - \frac{m_0}{r^3}$$

$$\Psi_3 = -\frac{3\, \sqrt{2}\epsilon \kappa_0 q_0^2 e^x}{2\, r^3 u} + \frac{\sqrt{2}\kappa_0 q_0^2 e^x}{4\, r^4}$$

$$\Psi_4 = -\frac{\epsilon \kappa_0 q_0^2 e^x}{r^3 u} + \frac{\kappa_0 q_0^2 e^x}{8\, r^4}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"